Title: Smart Student Management System Using Singly, Doubly and Circular Linked Lists

1.Problem Statement

In educational institutions, managing student records manually can be time-consuming and difficult. Searching, updating, and deleting student information becomes challenging when the number of students increases.

There is a need for a system that can efficiently manage student records, provide forward and backward navigation, schedule student turns for presentations and practicals, and analyze academic performance.

Therefore, the Smart Student Management System is developed using C++ and three types of linked lists: Singly Linked List, Doubly Linked List, and Circular Linked List. The system also compares their time and space complexities to demonstrate their practical applications.

2.Objectives

- To develop a student management application using C++.
- To store and manage student records using a Singly Linked List.
- To provide forward and backward navigation using a Doubly Linked List.
- To schedule student turns using a Circular Linked List.
- To calculate student grades, class average, and highest marks.
- To compare the time and space complexities of different linked lists.
- To understand the practical applications of linked list data structures.

3.Algorithm

Step 1: Start the program.

Step 2: Display the main menu.

Step 3: Accept the user's choice.

Step 4: Perform the selected operation:

Add a student record.

Display all student records.

Search for a student by roll number.

Update student details.

Delete a student record.

Navigate forward and backward using a Doubly Linked List.

Schedule student turns using a Circular Linked List.

Calculate grades, class average, and highest marks.

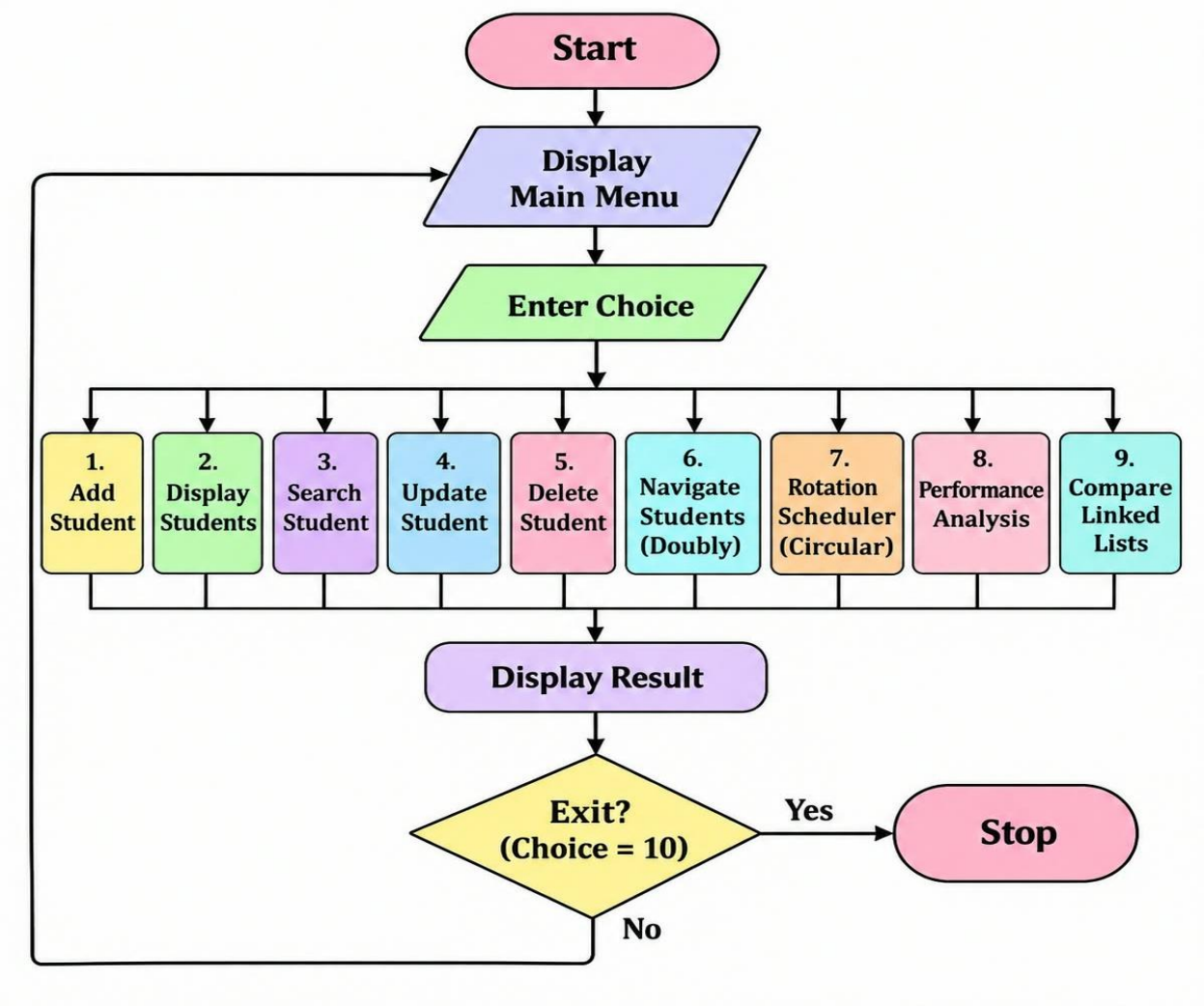
Compare linked lists with their time and space complexities.

Step 5: Display the result of the selected operation.

Step 6: Return to the main menu if the user has not selected Exit.

Step 7: Stop the program when the user selects Exit.

4.Flowchart



5.Conclusion

The Smart Student Management System successfully demonstrates the implementation of Singly Linked List, Doubly Linked List, and Circular Linked List using C++.

The application provides student record management, forward and backward navigation, circular student scheduling, and academic performance analysis. It also compares the time and space complexities of different linked list operations.

This project improves understanding of pointers, dynamic memory allocation, and linked list operations. In the future, features such as file handling, attendance management, and database connectivity can be added to improve the system.